Advanced Wireless Communications Laboratory Series

Robust Chirp Acquisition and Packet Recovery under Structured Interference

GNU Radio and Python simulation laboratory

Reproducible code, recorded measurements, and analysis

WC-LAB-04 2026

Executive summary

A receiver can detect a chirp preamble and still fail to recover the payload. This laboratory implements an end-to-end complex-baseband chirp link to measure both outcomes under structured interference. The transmitter sends four known chirps followed by 12 data-bearing chirp symbols. The receiver estimates the packet start, performs dechirping and FFT symbol decisions, reconstructs nine information bytes, and checks an appended CRC-8.

Four receiver paths process identical observations: unmodified samples, amplitude blanking, local spectral excision, and a combination of both. The experiment includes broadband barrage, swept-frequency, pulsed, and idealized event-triggered interference. At a desired-signal SNR of -7 dB, spectral excision increases packet recovery under swept interference at JSR $= 10$ dB from 38.3% to 98.0%. Under event-triggered interference at JSR $= 15$ dB, blanking increases recovery from 5.3% to 43.3%. Neither processing method improves all four interference cases. In particular, blanking reduces packet recovery at pulsed JSR $= 10$ dB from 9.7% to 4.3%.

The experiment contains 5,100 transmitted packets and 2,040 independent preamble-absent controls across 17 operating conditions and three seeds. Receiver comparisons are paired within each recording. All timing and payload outcomes are saved at packet level. This is a laboratory chirp framing model, not a complete LoRa PHY or an over-the-air RF evaluation.

Contents

1 Introduction **2**

2 System architecture **2**

3 Packet and channel model **2**

4 Receiver algorithms **3**

5 Measurements and validation **4**

6 Results **5**

7 Implementation and scope **7**

8 Conclusion **8**

1 Introduction

Chirp acquisition is necessary but is not sufficient for packet reception. Interference can suppress a preamble correlation peak, produce an incorrect timing lock, or corrupt data symbols after correct synchronization. A receiver comparison must distinguish these failures.

Ghanaatian et al. analyze digital LoRa reception and synchronization [1]. Xhonneux et al. investigate interference-aware multi-user synchronization and payload detection [2]. Barros and Studer address time-of-arrival estimation under interference and multipath using spatial processing [3]. Filtering-based chirp detection in the presence of narrowband interference is also studied by Maleki et al. [4]. These studies motivate the receiver questions considered here; their synchronization, spatial filtering, and multi-user detection algorithms are not implemented in this laboratory.

The present work compares two simple interference operations before an otherwise identical packet receiver. Its purpose is to establish where those operations improve measured recovery and where they remove useful signal information. No adaptive strategy is selected using the transmitted bits or the true packet arrival time.

2 System architecture

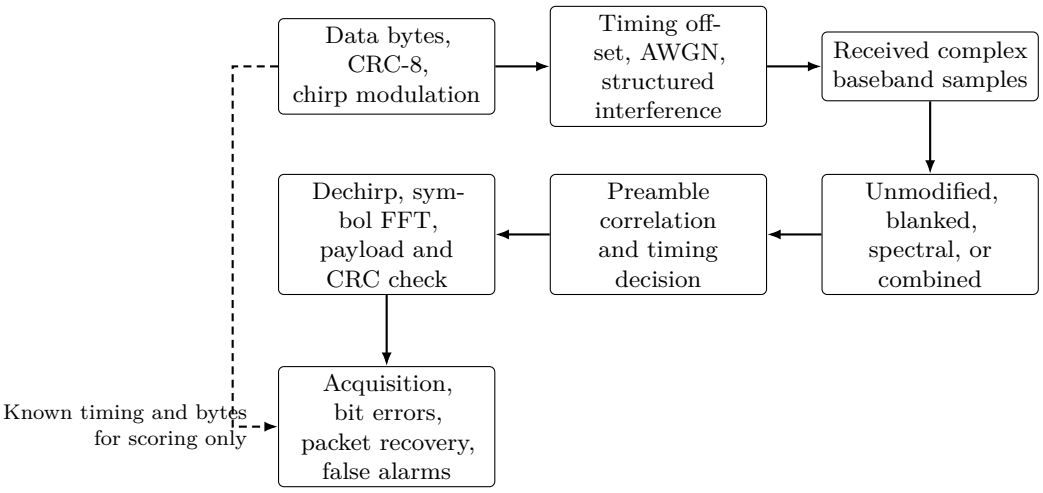


Figure 1: Four paired receiver paths share the same transmitted packet, noise, and interference realization. Only the receiver's sample stream is used for synchronization and demodulation; known timing and information bytes are used after decoding to score the output.

3 Packet and channel model

3.1 Chirp modulation and framing

The baseband upchirp has $N = 128$ complex samples:

$$c[n] = \exp\left(j\pi \left[\frac{n^2}{N} - n\right]\right), \quad 0 \le n < N. \tag{1}$$

An integer symbol $k \in \{0, \ldots, N-1\}$ is transmitted as

$$x_k[n] = c[n] \exp(j2\pi kn/N). \tag{2}$$

Four copies of $c[n]$ form the known preamble. Nine information bytes and a CRC-8 byte (polynomial 0x07, initial register zero, MSB-first) produce 80 protected bits. These bits are partitioned into 12 seven-bit symbols; the final four padding bits are zero. The frame therefore has $4N + 12N = 2,048$ samples. The CRC detects many errors but does not correct them. The packet does not include the standardized LoRa header, interleaver, whitening, or forward-error-correction operations.

An observation has 2,240 samples at $F_s = 125$ ksample/s (17.920 ms). The integer packet start is drawn uniformly from samples 32 through 160; the receiver searches all 193 admissible full-packet starts from 0 through 192. The nominal active-frame sample power is unity.

3.2 Channel and interference

The received observation is

$$r[n] = e^{j\phi} s[n - \tau] + w[n] + \sqrt{P_j}\, v[n], \qquad w[n] \sim \mathcal{CN}(0, \sigma_w^2), \tag{3}$$

where s is zero outside the packet, ϕ is an unknown constant phase, and v is a unit-mean-power record-length interference waveform. SNR is $10 \log_{10}(1/\sigma_w^2) = -7$ dB. The stated JSR is record averaged:

$$\mathrm{JSR} = 10 \log_{10} \left(\frac{\frac{1}{2240} \sum_{n=0}^{2239} |\sqrt{P_j} v[n]|^2}{1} \right). \tag{4}$$

For pulsed waveforms, instantaneous interference power during the active interval exceeds the record average. The tested JSR values are 0, 5, 10, and 15 dB.

Table 1: Synthetic interference conditions.

Class	Applied waveform
Barrage	Independent circular complex Gaussian samples over the entire record.
Swept frequency	Constant-envelope sinusoid with a randomized start frequency and a linearly changing instantaneous frequency across the record.
Pulsed	Gaussian burst occupying one random contiguous interval of 20% of the record.
Event-triggered	Gaussian burst starting 64 samples after the simulated packet arrival and lasting 384 samples. The onset is supplied by the simulator, not estimated by a reactive transmitter.

4 Receiver algorithms

4.1 Interference processing

All four receiver paths use fixed, documented settings. The baseline passes samples unchanged. The blanking path removes unusually large-amplitude samples using a robust record-level reference:

$$y_B[n] = r[n]\, \mathbf{1}\{|r[n]| \le 2.7 \,\mathrm{median}_m\, |r[m]|\}. \tag{5}$$

Spectral excision partitions the record into 35 disjoint 64-sample blocks. For each block b, let $R_b[k]$ be its 64-point DFT and let $Q_b[k] = |R_b[k]|^2$. Let k_b^* index the largest bin. It is excised, along with its two immediate neighbors, only if

$$Q_b[k_b^*] > 14 \,\mathrm{median}_k\, Q_b[k] \quad \text{and} \quad Q_b[k_b^*] > 0.18 \sum_k Q_b[k]. \tag{6}$$

A 64-point inverse DFT reconstructs each block after the selected bins have been set to zero. The combined receiver applies blanking first, then spectral excision. Neither operation uses the true onset or transmitted payload. The thresholds are fixed for all recorded experiments, so the reported changes do not include tuning to a particular test realization.

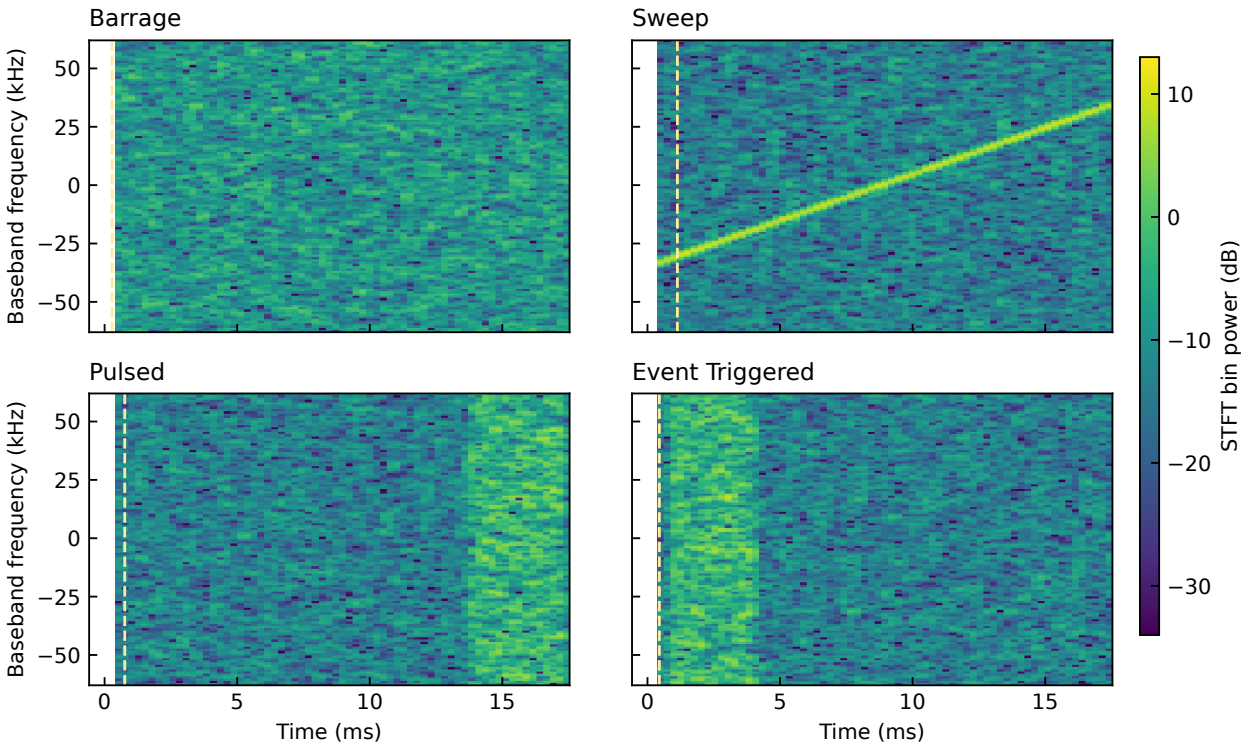


Figure 2: Received time-frequency examples at JSR = 10 dB. Dashed lines mark the true packet start; the same scale is used in all four panels. These are synthetic observations, not hardware spectrum measurements.

4.2 Acquisition and packet decisions

For each candidate start t, a normalized coherent matched filter searches the entire four-chirp preamble:

$$T(t) = \frac{\left|\sum_{n=0}^{4N-1} y[t+n]c^*[n \bmod N]\right|^2}{4N \sum_{n=0}^{4N-1} |y[t+n]|^2}. \tag{7}$$

For receiver m, $\hat{\tau}_m = \arg\max_{0 \le t \le 192} T_m(t)$ is accepted only when the maximum exceeds γ_m. Each γ_m is calibrated separately as the empirical 99th percentile of the maximum statistic from 1,500 independent noise-only records. The resulting nominal noise-only false alarm rate is 1%; this is not a constant-false-alarm guarantee when interference is present.

For an accepted timing estimate, symbol ℓ is extracted starting at $\hat{\tau}_m + 4N + \ell N$. Dechirping and an N-point DFT produce

$$Y_\ell[k] = \sum_{n=0}^{N-1} y[\hat{\tau}_m + 4N + \ell N + n]c^*[n]e^{-j2\pi kn/N}, \qquad \hat{k}_\ell = \arg\max_k |Y_\ell[k]|^2. \tag{8}$$

The 12 detected symbols are unpacked into 80 bits and checked using CRC-8. This is a noncoherent symbol decision with no channel estimate or soft decoder. The receiver cannot use the transmitted payload to correct a CRC failure.

5 Measurements and validation

The experimental design has 17 conditions: one noise-only reference and the 16 combinations of four interference classes and four JSR levels. Every condition is evaluated with 100 independent transmitted

packets and 40 preamble-absent controls for each of seeds 2026, 2027, and 2028. Thus each plotted point contains 300 paired packet observations; the full study contains 5,100 packets and 2,040 preamble-absent recordings. All four receivers operate on each observation.

A correct acquisition has $|\hat{\tau} - \tau| \leq 2$ samples. A threshold miss means no estimate was declared; a false lock passes the threshold outside that tolerance. These three events partition all preamble-present trials. The conditional bit-error rate counts incorrect information bits only for correctly acquired packets. Packet recovery requires a correct timing lock, CRC acceptance, and an exact match to the nine transmitted information bytes. The transmitted bytes are used only by the scoring script. A CRC pass without correct bytes is counted separately as an undetected erroneous acceptance. A preamble-absent false alarm is measured on independent control recordings.

Table 2: Experimental parameters and archived sample counts.

Parameter	Value
Sampling frequency; chirp length	125 kHz; 128 samples
Frame; observation	2,048; 2,240 complex samples
Information bytes; CRC	9; CRC-8 polynomial 0x07
Timing search; correct-lock tolerance	193 candidates; ±2 samples
SNR; JSR levels	−7 dB; 0, 5, 10, 15 dB
Threshold calibration	1,500 noise-only records; 1% target P_{FA}
Test repetitions	3 seeds; 100 packets plus 40 controls per condition
Packet study	5,100 transmitted; 2,040 absent controls
Mitigation DFT; amplitude multiplier	64 samples; 2.7× record median

6 Results

6.1 Packet recovery across interference levels

Table 3: Recovered packets out of 300 paired observations per condition.

Condition	Unmodified	Blanking	Spectral	Combined
No interference	300	300	300	300
Swept, JSR 10 dB	115	113	294	294
Swept, JSR 15 dB	0	0	216	216
Event-triggered, JSR 15 dB	16	130	16	130
Pulsed, JSR 10 dB	29	13	29	13
Barrage, JSR 10 dB	33	30	33	30

Spectral excision is effective against the slowly swept interferer in this model. At JSR = 10 dB, it recovers 294 of 300 packets, compared with 115 for the unmodified path. At 15 dB, it recovers 216 packets while the unmodified path recovers none. The 64-sample FFT finds concentrated energy that can be removed without deleting the entire chirp bandwidth.

For the event-triggered burst at 15 dB, blanking recovers 130 packets versus 16 without processing. Here the main effect is at acquisition: the strong burst obscures the preamble correlation. Blanking sacrifices samples but allows the remaining preamble to provide a more reliable timing peak. Neither operation improves broadband barrage performance. The 20%-duration pulsed noise also remains damaging; at 10 dB, blanking decreases successful packet recovery from 29 to 13 of 300. Removing high-amplitude samples does not reconstruct lost data symbols.

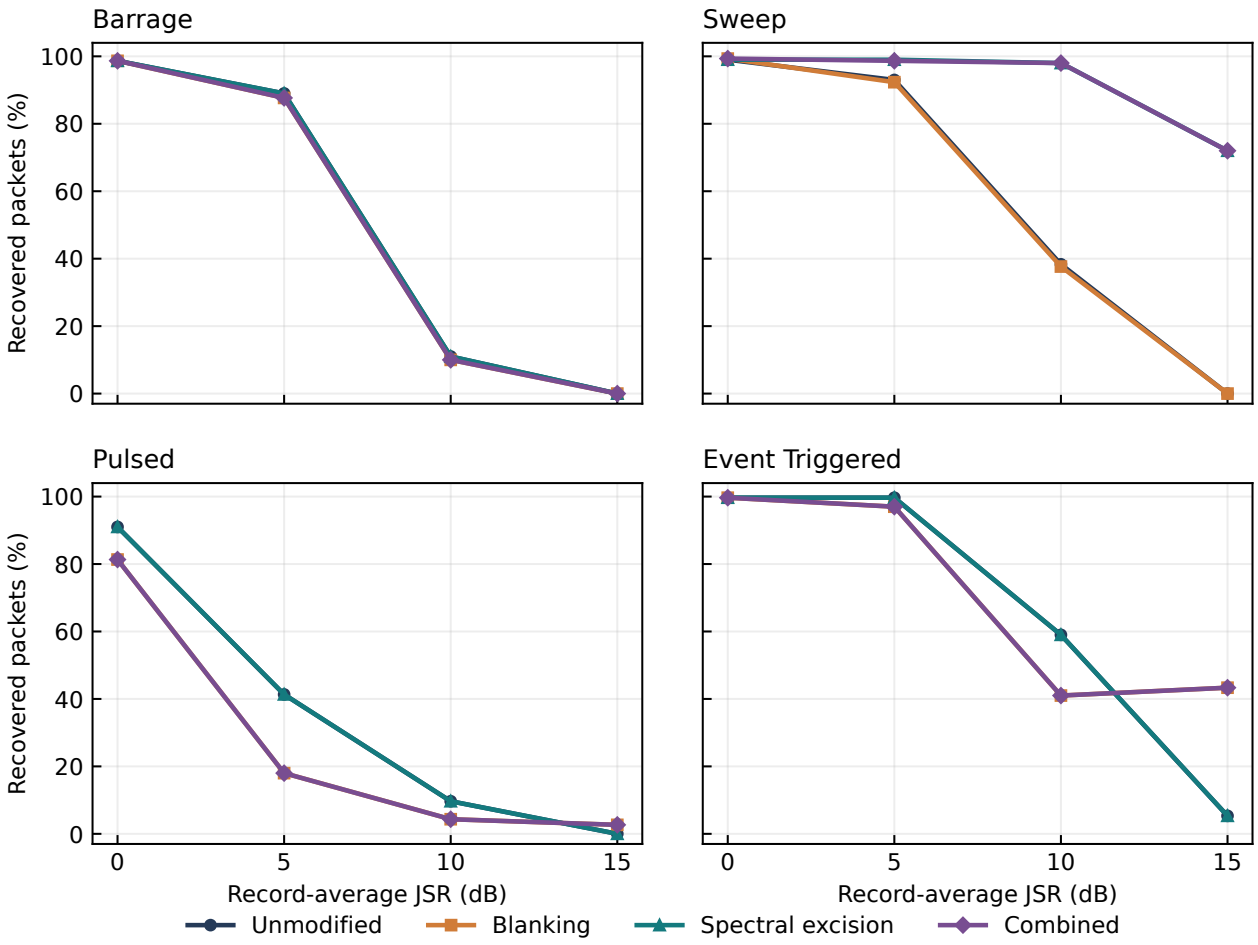


Figure 3: Correct payload recovery versus record-average JSR at SNR = −7 dB. All receiver paths operate on the same 300 packet observations at every point. Recovery requires a correct timing lock, valid CRC, and correct payload bytes.

6.2 Why acquisition and recovery differ

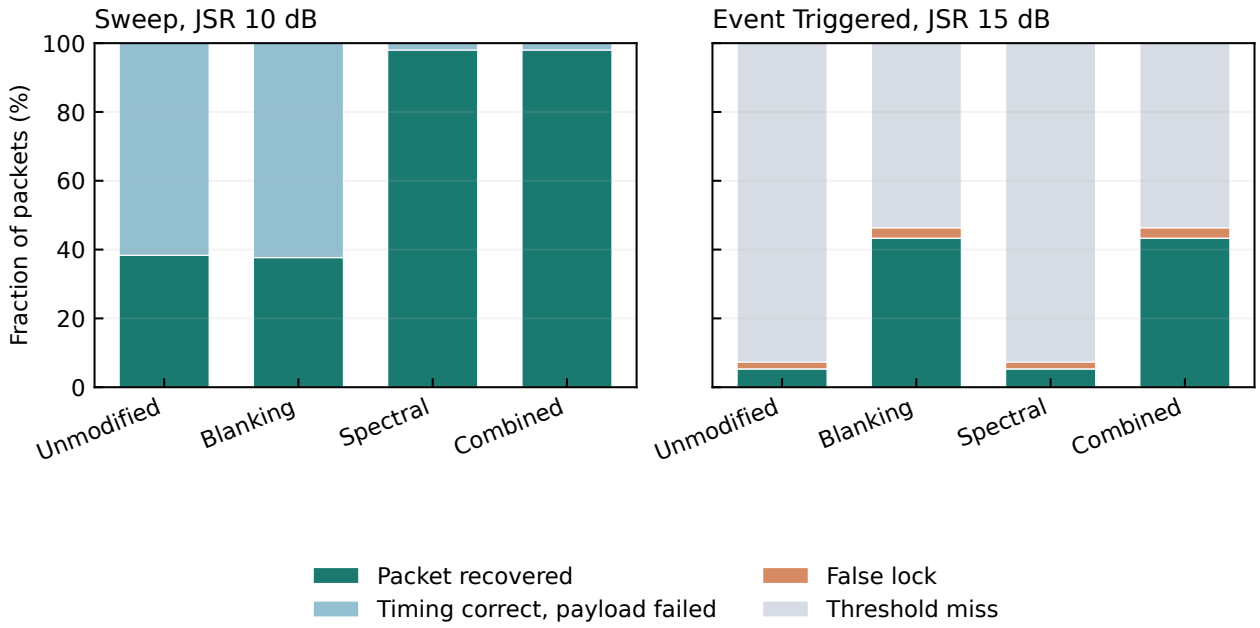


Figure 4: Mutually exclusive outcomes at two selected interference settings. Correct acquisition may still produce a corrupted payload; false locks and threshold misses are scored separately. Each bar contains 300 packets.

Under swept interference at JSR = 10 dB, every receiver correctly finds the preamble start in all 300 trials. Yet the unmodified receiver recovers only 115 packets. This is payload corruption, not acquisition failure. The same experiment shows why correct timing alone is an insufficient measure of interference robustness.

Under event-triggered interference at 15 dB, the unmodified receiver makes 278 threshold misses and six false locks. Blanking reduces threshold misses to 161, with nine false locks; the remaining correctly acquired packets are recovered. A lock that passes the threshold is not necessarily at the true packet start.

6.3 Effect on a data-symbol decision

The example shows the operation that accounts for the swept-interference improvement. A narrow frequency component can dominate a dechirped FFT decision even after synchronization succeeds. Local spectral excision changes that decision without relying on the transmitted symbol value. Such suppression can also remove desired-signal energy, explaining why it is not uniformly beneficial.

7 Implementation and scope

The Python simulation generates the transmitted chirps, independent Gaussian noise, and four interference models. It records packet timing, decoded bytes, and trial-level measurements. The GNU Radio Companion flowgraph uses an Embedded Python Block and provides interactive waterfall, spectrum, and packet-recovery displays. The reported probabilities are calculated from the packet-level CSV data.

All methods use the same coherent preamble search and 12 payload FFTs. Blanking adds a record-median and 2,240 amplitude comparisons. Spectral excision adds 35 forward and 35 inverse 64-point

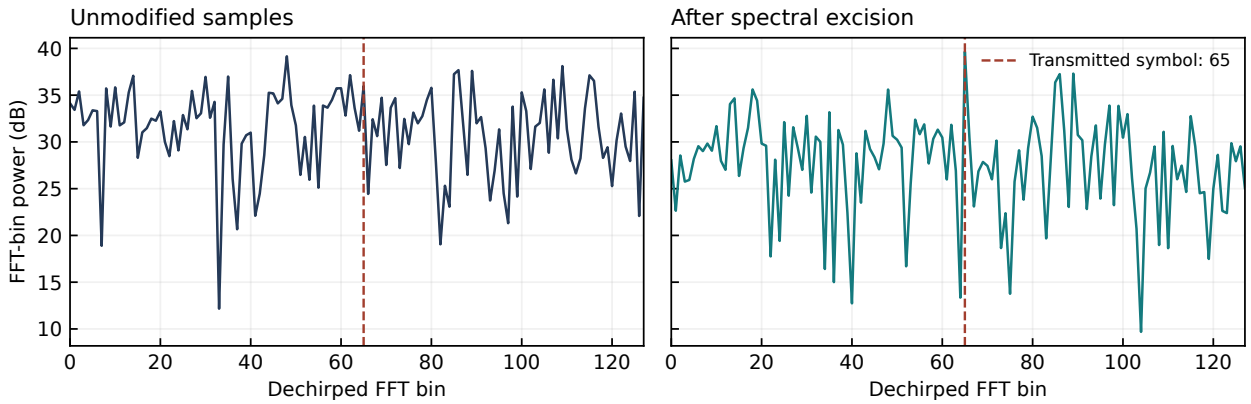


Figure 5: One selected swept-interference packet, JSR = 10 dB. The dashed line is the transmitted symbol bin (65). Spectral excision changes the largest dechirped FFT response from an incorrect bin to the transmitted bin. This is one deterministic illustrative packet, not an average result.

DFTs, with at most three frequency bins suppressed per flagged block. These counts describe the processing required by each method.

The model assumes a single complex receive channel, an integer timing offset, constant carrier phase, no carrier-frequency offset, no multipath, and no forward error correction. The event-triggered waveform uses simulator-known arrival time to schedule interference, not a physically modeled reactive transmitter. The packet format is a laboratory chirp model rather than a complete standardized LoRa physical layer.

8 Conclusion

This implementation extends chirp synchronization to an end-to-end packet recovery test. Under the specified model, short-time spectral excision substantially improves recovery against swept-frequency interference, and amplitude blanking improves acquisition during strong event-triggered bursts. Both methods can fail or reduce performance under other interference structures. The decisive result is not a universal best filter: receiver performance depends on how interference is distributed in time and frequency, and acquisition probability alone does not predict payload recovery.

References

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